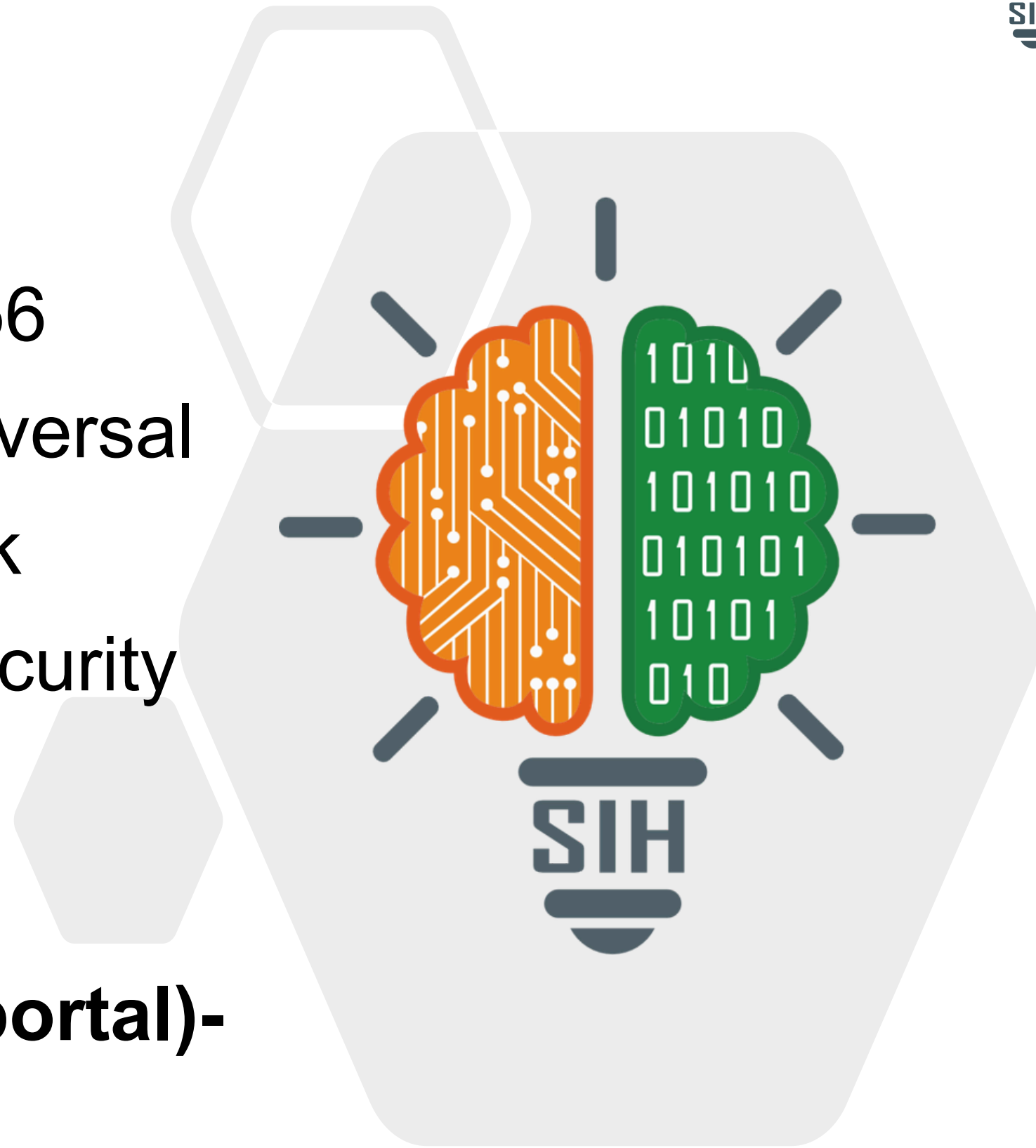


- **Problem Statement ID – 26156**
- **Problem Statement Title-** Universal Log Pre-processing Framework
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-**
Loggers





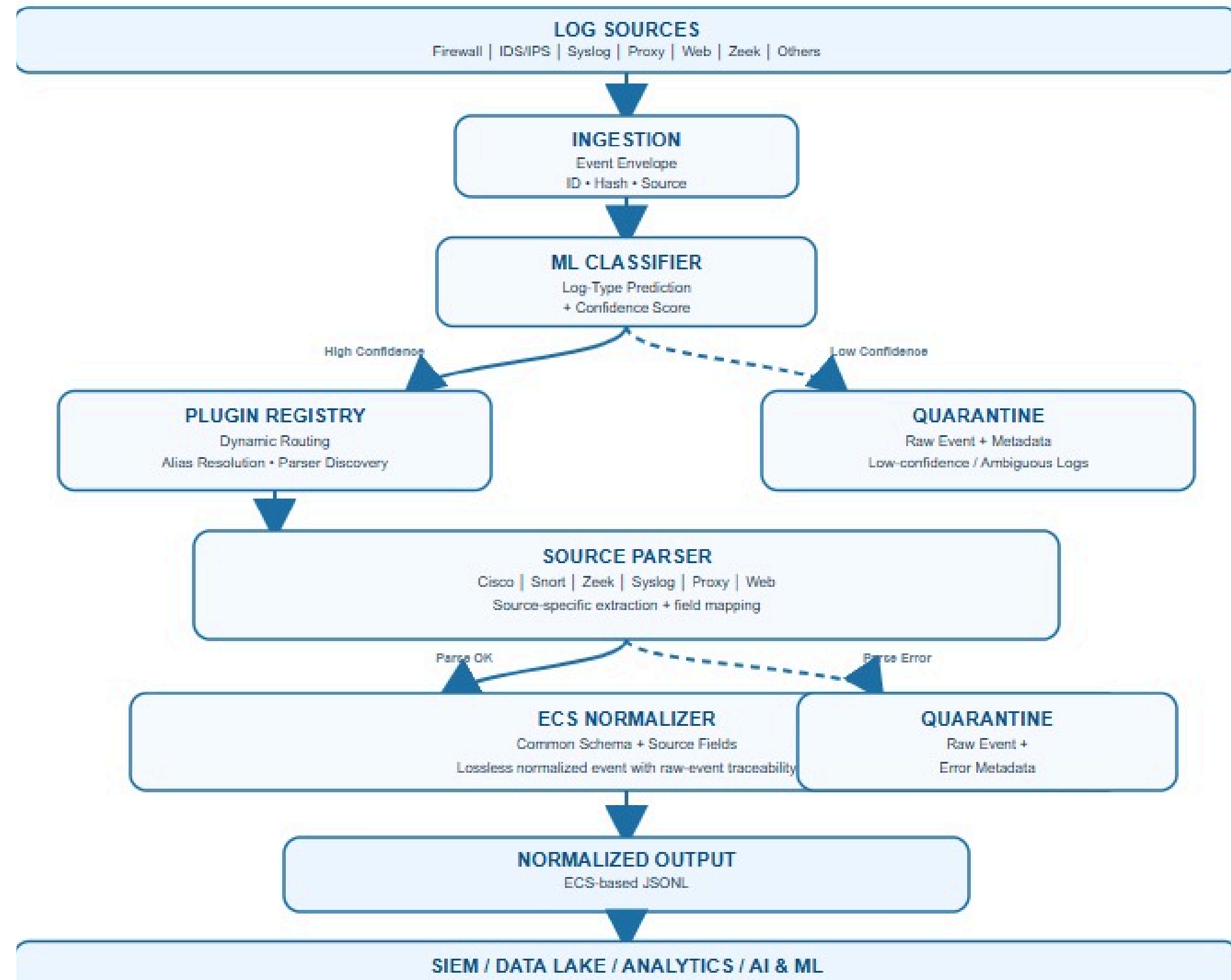
Problem:

Enterprise logs come in many formats such as Syslog, JSON, CEF-like text, and vendor-specific firewall or IDS formats.

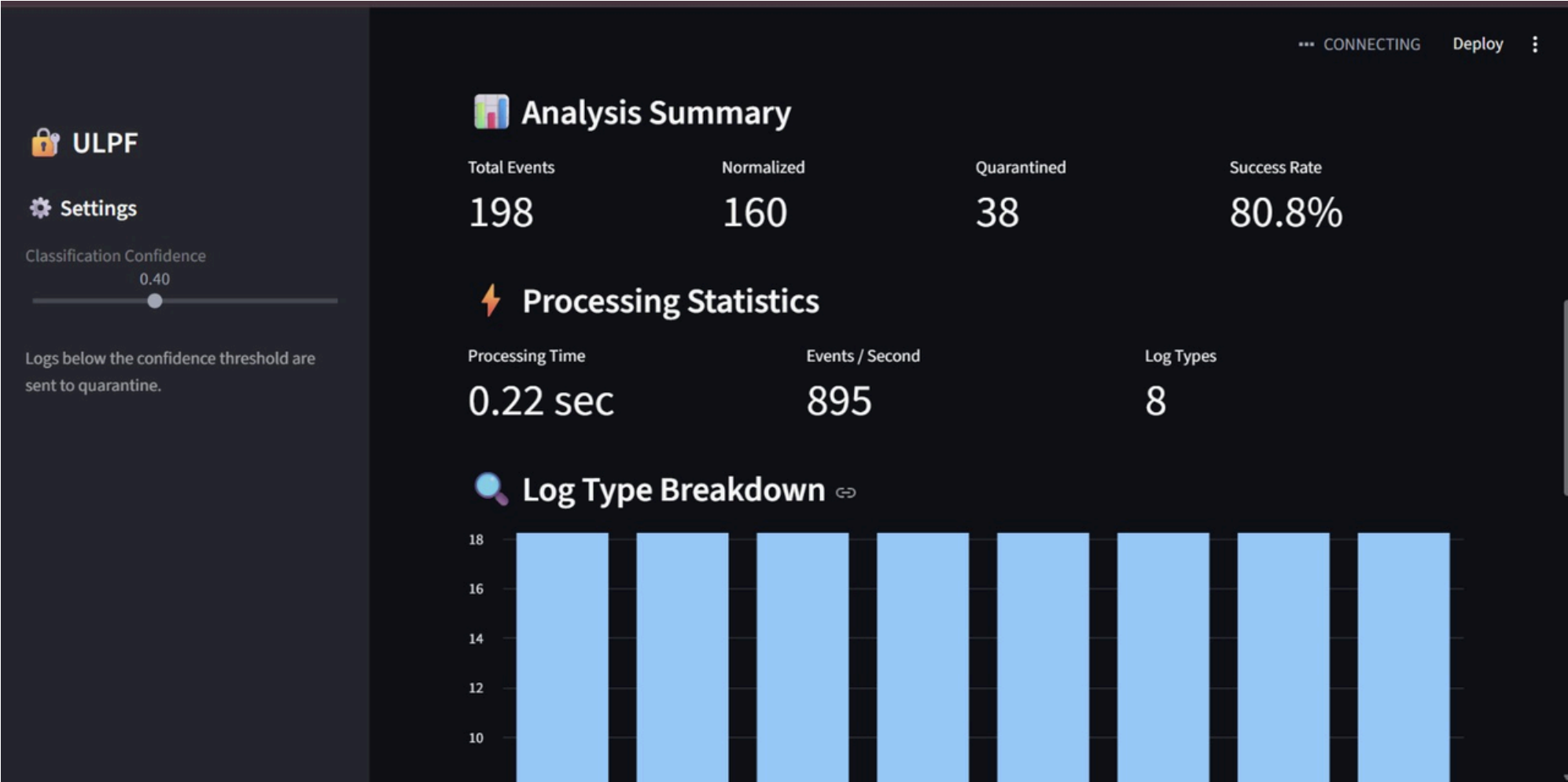
The Solution: ULPF

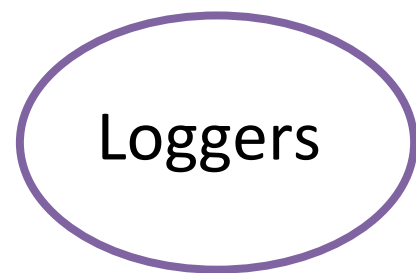
An air-gapped, lossless preprocessing layer that:

- Identifies & Routes: Uses ML classification to target the correct parser.
- Preserves Raw Data: Guarantees forensic losslessness by storing unaltered payloads in event.original.
- Standardizes to ECS: Maps vendor attributes into a common taxonomy (source.*, destination.*, user.*).
- Outputs Structured JSON: Feeds analytics-ready data directly to SIEM, data lakes, and ML pipelines.

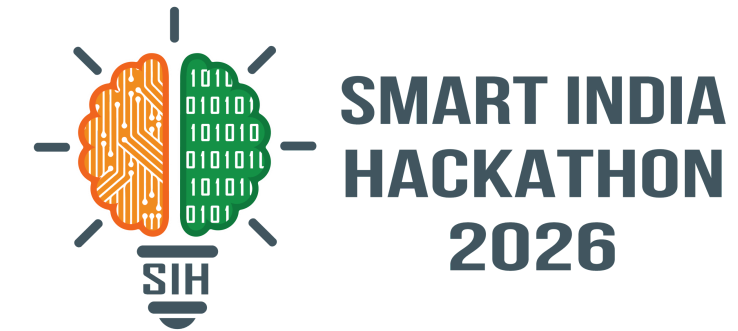


Layer	Technology
Language	Python 3.10+
ML & Classification	Sckit-Learn, TF-IDF and Logistic Regression
Log Parsing	Python Regex and JSON Parsing
Standardization	Elastic Common Schema (ECS) 8.11
Model Persistence	Joblib
Deployment	Docker
Output	JSONL





FEASIBILITY AND VIABILITY



Feasibility

- **Proven & Cost-Effective:** Uses established open-source technologies and reusable parsing components, reducing development and licensing effort.
- **Universal Log Processing:** Standardizes heterogeneous logs into a consistent, analysis-ready ECS-based structure.
- **Easy Integration:** Structured JSONL output can feed SIEM, data lake, analytics, and ML pipelines.
- **Deployment Flexibility:** Self-contained Docker deployment supports local and air-gapped environments.
- **Scalable & Extensible:** New log formats can be onboarded by adding representative training data, a parser plugin, and retraining the classifier — without modifying the core pipeline.

Viability

- **Clear Industry Need:** Addresses growing log volumes, format diversity, and source-specific parser maintenance.
- **Scalable Adoption:** Suitable for small deployments today, with the stateless architecture allowing horizontal scaling for larger workloads.
- **Operational Value:** Reduces parser development effort, improves data consistency, and preserves raw events for investigation.
- **Continuous Extensibility:** New sources can be added through plugins and additional training data without redesigning the core framework.
- **Long-Term Potential:** Can evolve with streaming ingestion, additional integrations, richer ML classification, and broader log-source coverage.

Challenges

- **Handling highly diverse log formats-** Different vendors and systems produce heterogeneous structures and schemas.
- **Processing large-volume log streams-** Handling growing log volumes while maintaining processing performance.
- **Maintaining parsing accuracy -** Correctly identifying and parsing unfamiliar or ambiguous log formats.
- **Protecting sensitive log data -** Handling corrupted, incomplete, or unexpected log entries.
- **Managing malformed or missing data -** Ensuring logs can be processed securely, including in isolated environments.
- **Continuous Source Expansion -** Maintaining and extending parsers as new log sources and formats emerge.

For Security Teams

- Faster SIEM onboarding
- Unified visibility across diverse log sources
- Better threat hunting and correlation
- For Compliance / Forensics
- Raw logs are preserved
- Full traceability is maintained
- Supports audits and investigations

For Engineering Teams

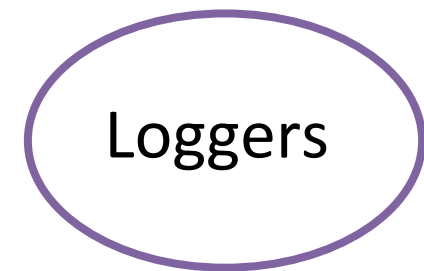
- Reduced parser development effort
- Easy plugin-based extension
- Cleaner separation of core logic and source-specific logic

For Analytics / AI

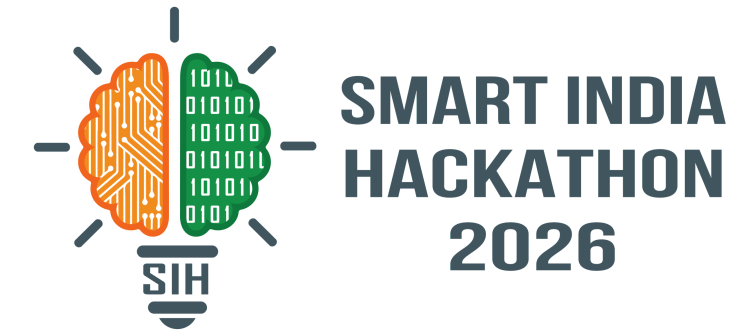
- ECS-like normalized events
- AI/ML-ready structured data
- Consistent schema across sources

Key impact statement

- ULPF reduces log-format chaos and turns raw enterprise telemetry into a universal, analytics-ready security data layer.



RESEARCH AND REFERENCES



Log Parsing & Normalization

Liu, Y., et al. (2022). "UniParser: A Unified Log Parser for Heterogeneous Log Data." Proceedings of the ACM Web Conference (WWW '22).

Addresses the challenge of generalizing log parsing across highly diverse log sources.

Zhang, T., et al. (2022). "System Log Parsing: A Survey." arXiv preprint.

Surveys existing log parsing approaches and highlights the challenges of converting heterogeneous raw logs into structured records.

Sapegin, A., et al. (2014). "Normalisation of Log Messages for Intrusion Detection." Journal of Information Assurance and Security, 9(3), 167–176.

Investigates common representations for heterogeneous security logs and proposes an extensible normalization format for intrusion detection.

Security Event Processing

Stroeh, K., Mauro Madeira, E. R., & Goldenstein, S. K. (2013). "An Approach to the Correlation of Security Events Based on Machine Learning Techniques." Journal of Internet Services and Applications, 4(7).

Presents collection, normalization and ML-based processing of security events from multiple sources.

Standards & Technical References

Elastic Common Schema (ECS) Documentation.

Common event schema used as the basis for ULPF's normalized event representation.

scikit-learn Documentation

Machine-learning framework used for TF-IDF feature extraction and Logistic Regression-based log-type classification.

Docker Documentation

Containerization reference used for packaging and deploying the ULPF runtime in an isolated environment.

Synthetic Security Telemetry

Cisco Talos (2026). "EvidenceForge: Synthetic Security Telemetry Generation Framework."

Used to generate realistic, multi-format synthetic security telemetry for ULPF testing and parser validation.